

Question		Answer	Mark	Mark scheme	Additional guidance
16	(a)	4.5	P1	for initial use of formula, eg $6 = 8k$	If figures are given as part of the answer they must be correct, but can allow ft
			P1	for a full process to find P_3 eg " $\frac{6}{8}$ " $\times 6$	
			A1	oe	
	(b)	Explanation	C1	for explanation Acceptable examples Yes, the population will increase as k is over 1 She is correct because $1.05 > 1$ Yes, each year the population will increase by 5% Yes, because $0.75 + 0.3 = 1.05$ Yes, a 0.3 increase is greater than the current 0.25 decrease Not acceptable examples Yes, the population will increase each year Yes, because there is an increase in k No because 0.3 is less than 0.75	